

BUSINESS MODEL & INVESTMENT PROSPECTUS

DISHARI

AI-Powered Smart Navigation System for the Visually Impaired

A low-cost, on-device tactile obstacle-detection platform bringing real-time spatial awareness to independent mobility for the blind and visually impaired.

Prepared by the Dishari Founding Team

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1. Executive Summary

Dishari is a low-cost, AI-powered wearable navigation aid that gives blind and visually impaired (BVI) users real-time, tactile awareness of obstacles in their walking path. Built on a Raspberry Pi 4B, a Pi camera, and an on-device YOLOv5-Nano object-detection model, Dishari divides the user's forward walking path into a 3x3 grid and converts detected obstacles into mechanical pop-up feedback delivered through nine independently actuated servo points — no earpiece, no audio dependency, and no reliance on cloud connectivity.

The global assistive-technology market for vision loss is large and structurally underserved by affordable hardware: the World Health Organization estimates that at least 2.2 billion people worldwide live with a vision impairment, and global cases of blindness rose from 33.4 million in 2000 to 43.3 million in 2020, with numbers continuing to climb as populations age. Existing electronic travel aids — such as smart canes and wearable cameras — are priced from several hundred to several thousand US dollars, placing them out of reach for the vast majority of users in low- and middle-income countries, where nine in ten people with vision loss live.

Dishari's core innovation is architectural: by running a lightweight, fully on-device inference pipeline on commodity single-board-computer hardware, we can deliver obstacle-awareness functionality at a fraction of the bill-of-materials cost of competing devices, with no recurring data or subscription dependency required for core safety functionality. We are seeking seed investment of \$180,000–\$250,000 to move from validated engineering prototype to certified pilot production, field-validated with partner institutions, and positioned for launch in South Asia within 18–24 months.

This document lays out the problem, our solution, the market opportunity, our business model, go-to-market strategy, financial projections, and the specific investment ask. All financial figures are illustrative planning estimates based on current bill-of-materials pricing and comparable-market benchmarks; they are intended as a starting point for investor diligence and will be refined jointly with investors and manufacturing partners.

2. The Problem

Independent mobility is one of the most persistent barriers to dignity, employment, and social participation for blind and visually impaired people. The traditional white cane, while indispensable, only detects obstacles at ground level and within arm's reach — it cannot warn of an oncoming cyclist, a parked vehicle, an animal, or an obstacle at torso height until the cane physically strikes it or the user is already at risk.

Electronic travel aids that go beyond the cane exist, but they largely serve high-income markets:

- Camera-based wearable readers (e.g., OrCam MyEye) are priced around \$4,250 — out of reach for the overwhelming majority of BVI individuals in developing economies.
- Smart canes (e.g., WeWALK) add ultrasonic obstacle detection and app-based navigation to a standard cane, typically priced from several hundred US dollars, plus optional AI subscription add-ons.
- Most solutions depend on smartphone pairing, cloud services, or audio feedback — which can be difficult to use in noisy environments, drains cognitive attention, or is culturally and linguistically limited in local markets.

The result is a large, growing population with a well-documented safety and independence need, and a market gap for a device that is affordable, robust, audio-independent, and deployable at scale through institutional and NGO channels rather than only high-income retail consumers.

3. The Solution — Dishari

Dishari captures a continuous video feed from a Raspberry Pi Camera, defines a bottom-middle region of interest representing the user's walking path, and divides it into a 3×3 grid. A YOLOv5-Nano object-detection model — selected specifically for its ability to run in real time on low-power embedded hardware — identifies high-priority obstacle classes (people, vehicles, animals, poles, furniture, bags, and similar walking-path hazards) while filtering out non-obstructive background such as sky or open road.

Each of the nine grid segments generates a binary obstacle signal that drives a dedicated servo motor via a PCA9685 driver, converting a rotary motion into a linear tactile pop-up. The user feels — rather than hears — exactly where in their path an obstacle lies, enabling faster, more intuitive spatial judgment without occupying the auditory channel that BVI users rely on for orientation and safety cues from their environment.

Key differentiators

- Fully on-device inference — no internet dependency for core safety functionality, critical for reliability in low-connectivity regions.
- Tactile-first feedback — preserves the user's hearing for environmental awareness, unlike audio-based competitors.
- Commodity hardware architecture — built on a Raspberry Pi 4B and off-the-shelf components, giving a materially lower bill-of-materials cost than proprietary competitor hardware.
- Modular, serviceable design — individual servo/segment units can be repaired or replaced in the field without returning the whole device.
- ~8–12 FPS real-time detection at 5V/3A power draw, suitable for a portable battery pack.

4. Market Opportunity

The scale of the addressable population is large and growing. According to WHO and the Global Burden of Disease study, at least 2.2 billion people worldwide have some form of vision impairment, and the global blind population rose from 33.4 million in 2000 to 43.3 million in 2020, with the number of people affected by vision loss projected to keep rising as populations age. Roughly 90% of people with vision loss live in low- and middle-income countries — precisely the segment underserved by today's premium-priced assistive devices.

Market sizing (illustrative)

Tier	Definition	Estimated population	Commentary
TAM	Global blind & severely visually impaired population	~45–50 million (blind, 2020 baseline, growing)	Full global addressable population; mobility-focused electronic tech
SAM	Blind population in South & Southeast Asia reachable via NGO, government-welfare, and school-for-the-blind institutional channels	~8–12 million (illustrative)	Region combines high prevalence, sensitivity, and an active NGO tech distribution ecosystem

Tier	Definition	Estimated population	Commentary
SOM	Bangladesh + neighboring pilot markets, institutional and direct-to-consumer, first 3 years	10,000–25,000 addressable users; initial target 1,000–2,000 units sold	Realistic first-mover footprint achievable through institutional partnerships and NGO procur

Table 1. Market sizing tiers. Figures are planning estimates for investor discussion, to be refined with primary market research.

Beyond individual end users, a significant share of near-term revenue is expected to come from institutional buyers — schools for the blind, disability welfare NGOs, government rehabilitation programs, and CSR-funded procurement — which reduces customer-acquisition cost relative to a pure consumer retail model and shortens the sales cycle through bulk procurement.

5. Business Model Canvas

Block	Details
Customer Segments	(1) Individual BVI consumers and their families; (2) Institutional buyers — schools for the blind, NGOs, rehabilitation centers; (3) Government social-welfare and disability-support programs; (4) Assistive-technology distributors and retailers
Value Proposition	Real-time, tactile, audio-independent obstacle awareness at a fraction of the cost of existing electronic travel aids, engineered for reliable operation in low-connectivity, resource-constrained environments
Channels	Direct sales to NGOs and institutions; partnerships with disability welfare organizations; assistive-technology distributors; university/CSR pilot programs; online and retail for individual consumers
Customer Relationships	Hands-on onboarding and orientation training (in partnership with mobility instructors); dedicated support line; community user feedback loop for iterative firmware/model improvements; annual maintenance contracts for institutional clients
Revenue Streams	Hardware unit sales (B2C and B2B); institutional bulk contracts; annual maintenance & extended warranty plans; grant and CSR-funded distribution; future licensing of the detection/feedback platform to mobility-aid manufacturers
Key Resources	Proprietary ROI-grid tactile feedback mechanism and firmware; trained detection model and dataset; engineering and embedded-systems talent; manufacturing and assembly partnerships
Key Activities	Embedded software and model development; hardware assembly and quality testing; field trials with BVI users; certification and compliance; partnership development with NGOs and government bodies
Key Partners	Component and contract-manufacturing suppliers; disability welfare NGOs and schools for the blind; university research labs; government assistive-technology procurement programs; impact investors and CSR foundations
Cost Structure	Bill-of-materials and assembly; R&D and firmware maintenance; certification/compliance; field testing and user training; sales & partnership development; customer support and warranty servicing

Table 2. Dishari Business Model Canvas.

6. Revenue Model & Pricing Strategy

Dishari's estimated bill of materials — Raspberry Pi 4B (8GB), Pi Camera v2, nine SG90-class servos, a PCA9685 driver, frame, harness, and power components — is projected in the range of \$180–\$230 per unit at low-volume assembly, with meaningful reductions expected at higher production volumes through component sourcing and localized assembly. This BOM sits well below the retail price of comparable electronic travel aids such as camera-based readers priced above \$4,000 and smart canes priced from several hundred dollars upward, giving Dishari room for a sustainable margin even at an accessible price point.

Proposed revenue streams

- Direct hardware sales (B2C): retail unit price target in the range of \$280–\$380, positioned well below premium competitors while sustaining healthy gross margin.
- Institutional / bulk contracts (B2B): discounted per-unit pricing for NGO, school, and government procurement at volume, with predictable multi-unit order sizes.
- Annual maintenance & extended warranty (AMC): a recurring service fee covering servo/battery replacement, firmware updates, and support — the primary recurring-revenue line.
- Grant- and CSR-subsidized distribution: units funded by development partners or corporate social responsibility budgets, distributed at no or reduced cost to end users while Dishari is paid at or near full unit economics by the funding partner.
- Platform licensing (future): licensing the ROI-grid detection-to-tactile-feedback architecture to established mobility-aid manufacturers once the core technology is field-proven, as a longer-horizon, higher-margin revenue line.

This blended model — hardware-led with an institutional and recurring-service overlay — is deliberately chosen to reduce dependence on individual retail affordability while keeping the end-user price point within reach for direct purchase where feasible.

7. Go-to-Market Strategy

Phase 1 — Validate (0–6 months)

- Finalize the ruggedized, field-ready hardware enclosure and harness design.
- Run structured field trials with a partner school for the blind and/or disability NGO in Bangladesh to gather usability data and safety validation.
- Incorporate mobility-instructor feedback into the tactile feedback mapping and grid calibration.

Phase 2 — Certify & Pilot Produce (6–12 months)

- Pursue relevant product safety and electrical compliance certification for target markets.
- Run a pilot production batch (target: 100–500 units) through a contract manufacturing/assembly partner.
- Establish 2–3 anchor institutional partnerships as reference customers and distribution channels.

Phase 3 — Launch & Scale (12–24 months)

- Commercial launch in Bangladesh via NGO and institutional channels, supplemented by direct-to-consumer online sales.
- Build out AMC/warranty servicing infrastructure and a local repair/support network.

- Begin regional expansion discussions across South Asia, leveraging reference deployments and impact data.

Phase 4 — Regional Expansion (24–36 months)

- Expand institutional and government-procurement relationships across additional South and Southeast Asian markets.
- Explore platform-licensing conversations with established assistive-technology manufacturers.
- Layer in companion-app features and outdoor/GPS navigation as an optional add-on tier.

8. Competitive Landscape

Solution	Feedback mode	Connectivity need	Illustrative price	Dishari's position
Traditional white cane	Physical/tactile (ground-level only)	None	Low cost	Dishari adds proactive, to obstacle awareness the ca cannot provide
Smart cane (e.g., WeWALK-type)	Vibration + app audio/voice	Smartphone pairing for full features	Several hundred USD+, plus optional AI subscription	Dishari is a fully on-device based tactile system with smartphone dependency function
Wearable AI camera reader (e.g., OrCam MyEye-type)	Audio narration	Largely on-device but premium priced	~\$4,250	Dishari targets obstacle-p at a small fraction of this p rather than object/text re
Dishari	Multi-point mechanical tactile pop-up across a 3x3 path grid	None required for core safety function	Target \$280–\$380 retail / discounted institutional pricing	Affordable, audio-indepen connectivity-independent built for path-obstacle aw

Table 3. Competitive positioning. Competitor pricing reflects publicly listed figures at time of writing and may vary by region and configuration.

9. Illustrative Financial Projections

The figures below are planning-stage estimates intended to frame unit economics and funding requirements for investor discussion. They are not audited projections and will be refined during due diligence with manufacturing and distribution partners.

Unit economics (illustrative, at scale)

Metric	Estimate
Bill of materials (BOM) per unit	\$180 – \$230
Assembly, QC & packaging per unit	\$20 – \$35
Target B2C retail price	\$280 – \$380
Target institutional/bulk price	\$220 – \$280
Estimated gross margin (blended)	25% – 35%

Metric	Estimate
Annual maintenance contract (per unit/year)	\$15 – \$25

3-year volume and revenue outlook (illustrative)

Year	Units (cumulative)	Primary channel focus	Illustrative revenue range (USD)
Year 1	150 – 300 (pilot & field trial units)	Field validation, NGO pilots, grant-subsidized units	\$40,000 – \$90,000
Year 2	1,000 – 2,000	Institutional contracts + early direct-to-consumer sales	\$250,000 – \$550,000
Year 3	3,000 – 6,000	Regional scale-up, AMC base growth, distributor partnerships	\$800,000 – \$1,800,000

Table 4. Illustrative 3-year volume and revenue outlook. To be validated jointly with investors against manufacturing capacity and channel-partner commitments.

10. Funding Ask & Use of Funds

We are seeking a seed investment of \$180,000–\$250,000 to move Dishari from validated engineering prototype to a certified, field-tested product ready for institutional pilot deployment and initial commercial launch.

Use of funds	Approx. allocation	Purpose
Product engineering & certification	30%	Enclosure design, ruggedization, safety and compliance certification
Pilot manufacturing run	25%	100–500 unit production batch through a contract assembly partner
Field trials & user research	15%	Structured trials with partner NGOs/schools for the blind; mobility-instructor collaboration
Team & operations	15%	Core engineering, assembly QC, and operations hires
Sales, partnerships & marketing	10%	Institutional partnership development, distributor outreach
Working capital / contingency	5%	Buffer for component price volatility and timeline risk

Table 5. Proposed use of funds. Allocation percentages are indicative and open to negotiation with investors.

11. Team

Dishari is developed by a founding team from the Department of Electrical & Electronic Engineering, Rajshahi University of Engineering & Technology (RUET), with direct hands-on experience across embedded systems, computer vision, and hardware prototyping:

- Md. Shafiqul Islam — Co-founder, embedded systems & hardware integration
- Gourakant — Co-founder, AI/computer vision and firmware development
- Miratul Mariam Rinte — Co-founder, systems design and testing
- Md. Nuhi-Alamin, Assistant Professor, Dept. of EEE, RUET — Faculty Supervisor and technical advisor

The team's engineering background is directly evidenced by working prototypes already built and tested, including real-time YOLOv5n obstacle detection on embedded hardware and a functioning 3×3 servo-driven tactile feedback array — substantially de-risking the core technical execution ahead of this funding round.

12. Key Risks & Mitigation

Risk	Mitigation
Hardware durability in real-world outdoor/monsoon conditions	Ruggedized enclosure design and field testing prioritized in Phase 1 before scale-up
User adoption / learning curve for a new tactile feedback system	Co-design and onboarding with mobility instructors and partner NGOs; iterative feedback-mapping refinement
Component price volatility / supply chain	Working capital buffer; supplier diversification; localized assembly partnerships
Certification and regulatory timelines	Early engagement with certification bodies budgeted into Phase 1–2 timeline
Competition from well-funded international players	Cost structure and institutional/NGO channel strategy targets an underserved price segment competitors do not efficiently serve

13. Closing Statement

Dishari addresses a large, well-documented, and growing global need with a working, field-tested technical approach and a cost structure purpose-built for the markets where the need is greatest. We are inviting investors who share a conviction that safe, independent mobility should not be a luxury good to partner with us in taking Dishari from validated prototype to a certified product reaching real users. We welcome the opportunity to walk investors through a live demonstration of the working prototype and to discuss this proposal in further detail.